

PROJECT TECHNICAL DOCUMENTATION

Secure Multi-Site Enterprise Architecture Design & Implementation

1. EXECUTIVE SUMMARY & CORE TECHNOLOGIES

This documentation specifies the deployment of a highly secure, scalable, and redundant multi-site enterprise network infrastructure. The architecture spans multiple regions (Cairo & Alexandria Headquarters) utilizing industry-standard routing, switching, and advanced security protocols to guarantee maximum uptime, traffic isolation, and secure data transit.

VLSM / FLSM	Inter-VLAN Routing (MLS & RoaS)	OSPFv2	EIGRP	Route Redistribution	Site-to-Site VPN	NAT / PAT
Extended ACLs	AAA / Authentication	Port Security	DHCP Snooping	SPAN Mirroring	SSH / Secure Telnet	

2. IP ADDRESSING & SUBNETTING SCHEMES

The network employs a hybridized addressing strategy. The primary operational zones utilize **Variable Length Subnet Masking (VLSM)** to optimize efficiency and prevent address wastage, whereas secondary regions deploy a uniform **Fixed Length Subnet Masking (FLSM)** schema.

Table 2.1: Zone A Address Space (10.0.0.0 VLSM)

ZONE / VLAN	NETWORK ID	SUBNET MASK	GATEWAY / RANGE	BROADCAST IP	TARGET HOSTS
MANAGE (VLAN 30)	10.0.0.0 /25	255.255.255.128	10.0.0.1 (10.0.0.2 - .126)	10.0.0.127	90 Hosts
HR (VLAN 20)	10.0.0.128 /26	255.255.255.192	10.0.0.129 (10.0.0.130 - .190)	10.0.0.191	50 Hosts
IT (VLAN 10)	10.0.0.192 /26	255.255.255.192	10.0.0.193 (10.0.0.144 - .254)	10.0.0.255	40 Hosts
ACC (VLAN 40)	10.0.1.144 /28	255.255.255.240	10.0.1.145 (10.0.1.146 - .158)	10.0.1.159	10 Hosts
SERVER (VLAN 50)	10.0.1.160 /28	255.255.255.240	10.0.1.161 (10.0.1.162 - .174)	10.0.1.175	8 Hosts
Infrastructure Links	10.0.1.192/196/200/204 / 30	255.255.255.252	Point-to-Point WAN Connections	Multi-cast	Point-to-Point

Table 2.2: Zone B Address Space (20.0.0.0 VLSM)

ZONE / VLAN	NETWORK ID	SUBNET MASK	USABLE RANGE	BROADCAST IP	TARGET HOSTS
ACC (VLAN 14)	20.0.0.0 /26	255.255.255.192	20.0.0.1 - 20.0.0.62	20.0.0.63	33 Hosts
MANAGE (VLAN 13)	20.0.0.64 /27	255.255.255.224	20.0.0.65 - 20.0.0.94	20.0.0.95	22 Hosts
HR (VLAN 12)	20.0.0.96 /27	255.255.255.224	20.0.0.97 - 20.0.0.126	20.0.0.127	16 Hosts
IT (VLAN 11)	20.0.0.128 /28	255.255.255.240	20.0.0.129 - 20.0.0.142	20.0.0.143	12 Hosts
SERVER (VLAN 15)	20.0.0.144 /28	255.255.255.240	20.0.0.145 - 20.0.0.158	20.0.0.159	8 Hosts

Table 2.3: Branch Remote Sites (192.168.1.0 FLSM)

DEVICE ID	ASSIGNED VLAN	NETWORK ID	SUBNET MASK	USABLE HOST RANGE	BROADCAST IP
Switch4	VLAN 16	192.168.1.0 /27	255.255.255.224	192.168.1.1 - 192.168.1.30	192.168.1.31
Switch3	VLAN 17	192.168.1.32 /27	255.255.255.224	192.168.1.33 - 192.168.1.62	192.168.1.63
Switch2	VLAN 18	192.168.1.64 /27	255.255.255.224	192.168.1.65 - 192.168.1.94	192.168.1.95
Switch0	VLAN 19	192.168.1.96 /27	255.255.255.224	192.168.1.97 - 192.168.1.126	192.168.1.127

3. ROUTING, CORE SWITCHING & INTER-VLAN TOPOLOGIES

The routing matrix is split into diverse autonomous architectures optimized for local convergence speeds and stability, bound together through granular integration mechanisms:

Core Dynamic Routing Domain

- **OSPF Core Backbone:** Deployed inside Core Area 0 for lightning-fast SPF convergence and link-state structural mapping.
- **EIGRP Regional Edge:** Implemented at remote modules utilizing composite metrics (bandwidth/delay) for maximum stability.
- **Bi-directional Redistribution:** Executed meticulously at boundary routers using customized metric definitions to cross-populate network topologies flawlessly.

Layer 3 Layer Switching & VLANs

- **Multi-Layer Switch (MLS) Routing:** core VLAN interfaces are assigned SVI IPs on 3560/3850 hardware blocks executing dynamic hardware-level packet forwarding.
- **Router-on-a-Stick (RoaS):** Applied on branch router interfaces utilizing 802.1Q sub-interfacing encapsulation tags to bridge low-density remote VLAN segments.

4. EDGE APPLICATION SERVICES

Dynamic Addressing Infrastructure

- **DHCP Servers:** Deployed inside the primary data rooms serving automated IP allocations directly to local VLAN clients.
- **DHCP Router Pools:** Configured natively on peripheral branch edge devices using `ip helper-address` relays to dynamically map IP pools to distinct network nodes.

Corporate App & Resolution Layer

- **Dual-Site HTTP Web Services:** Hosts independent localized web servers providing `cairo.corp` and `alex.corp` intranets.
- **Centralized DNS Services:** Provides local domain name resolution resolving canonical client queries to internal web resources.

5. BOUNDARY ENGINEERING & SITE-TO-SITE SECURITY

To preserve WAN confidentiality, all inter-site transport channels are wrapped in strong cryptographic mechanisms across public routing realms:

- **Site-to-Site IPsec VPN:** Secure tunnels are bound over public networks connecting Cairo and Alexandria sites using Crypto maps, IKE Phase 1 (ISAKMP) policies, and IKE Phase 2 ESP transform sets for data integrity and encryption.
- **NAT / PAT (Port Address Translation):** Masks internal non-routable private addresses behind single public IP pools to allow seamless public internet transit.
- **Extended Access Control Lists (ACLs):** Strict layer-3 and layer-4 filtering constraints are deployed at local boundary interfaces to selectively isolate critical operations, blocking non-essential host inspection protocols.

6. LAYER-2 HARDENING & SECURITY IMPLEMENTATION

Securing Infrastructure Components: Mitigation vectors against local network vulnerability exploitation.

1. **Port Security:** Implemented on access switches to map and restrict maximum valid MAC addresses per interface, blocking frame-flooding attacks and sending shut-down triggers to unauthorized nodes.
2. **DHCP Snooping:** Establishes trusted vs. untrusted physical boundary ports on local switches, inspecting DHCP traffic to drop rogue DHCP server responses and poisoning attempts.
3. **AAA & Authentication Architecture:** Deployed via centralized TACACS+/RADIUS infrastructure alongside local fallback configurations to thoroughly manage administrative permissions.
4. **Secure Management Channels (SSH):** All standard telnet interfaces have been structurally disabled on infrastructure devices in favor of cryptographic **SSHv2** protocols.
5. **IT Management Override:** Enforces access-class filters across virtual terminal lines (VTY), permitting exclusively the designated IT Manager workstation IP to gain Telnet/SSH control over peripheral switches.
6. **Switched Port Analyzer (SPAN):** Configured on core routing nodes to actively mirror live ingress and egress packet lines, forwarding telemetry directly to the IT Manager's SPAN analytics server for traffic profiling.